COMPUTER VISION - ITA0512

CLASS TEST

Yavanika.S
192521258

24/25

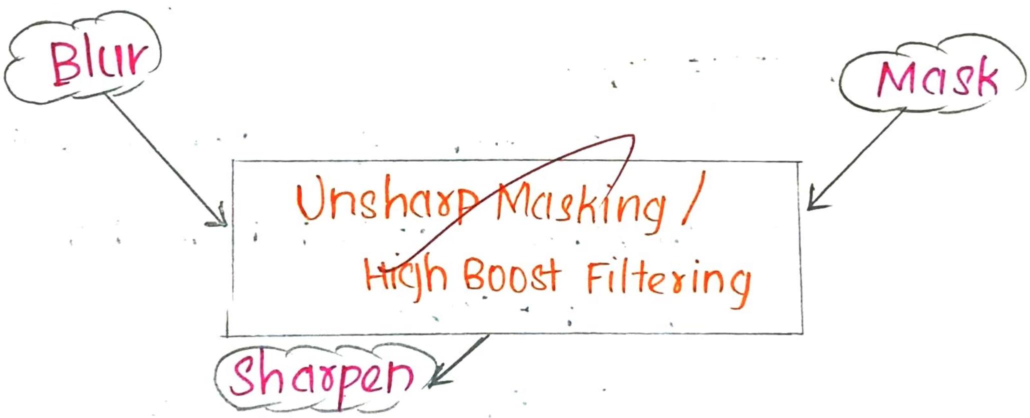
Good

Q.1 Edge Enhancement Without Amplifying Noise significantly:

Plain Edge sharpening [eg. Laplacian alone] boosts noise along with edges, because noises also have high frequency, sharp Intensity changes.

"Unsharp Masking" solves this by smoothening first, then adding back only the useful Edge detail.

⇒ It works in 3 steps



1. Blur:

Apply Gaussian Filter to original image to get a smoothed version.

2. Mask:

Subtract the blurred Image from the original.

$\boxed{\text{Mask} = \text{original} - \text{Blurred}}$

3. Sharpen:

Add the Mask back to original with a weight k.

$$\boxed{\text{Sharpened} = \text{Original} + K \times \text{Mask}}$$

$$\left\{ g(x,y) = f(x,y) + k\,[f(x,y) - f_{blur}(x,y)], \quad k >= 1 \right\}$$

(high boost)

Original Image → Gaussian Blur → Mask = Original − Blurred

→ Sharpened = Original + k(Mask) → Enhanced Edges (Low noises)

JUSTIFICATION:

⇒ Because the Mask is built from a smoothed copy, random noise is not re-amplified.

⇒ While True edges gets Boosted.

Expected Result:

Crisper Edges and outlines, with noise staying visibly low.

2. **Poor seperation between object background intensities**

Method : Otsu's Thresholding Method.

⇒ A single fixed threshold chosen by guesswork ofteen fails when object and background grey levels overlap.

⇒ Otsu's Method picks the threshold automatically and optimally from the images own Histogram.

It occurs in three steps:

Otsu's Thresholding Method → Histogram

Otsu's Thresholding Method → Test all Threshold

Otsu's Thresholding Method → Pick the best t.

Grayscale Image → Compute Histogram → Test each t. between class variance

→ Pick t with Max variance → Apply threshold segmentation.

In "otsu's Thresholding Method", Using Grayscale Image

⇒ compute Histogram.

Then, we test each t between class variance and Pick t with Max variance.

P1 ⇒ Finally Applying Threshold segmentation.

Expected Result: A clean binary Image with the object properly seperated from background.

3. **Multiple Regions with similar Intensity but different Texture;**

- Region based segmentation:

If Regions have same Average Intensity - based Methods [Like threshold] cannot tell them Apart.

⇒ Region growing can still seperate them if we grow Regions using Texture features, (eg: Local variance) Instead of Plain Intensity.

The steps are:

⇒ pick seed pixel

⇒ check neighbour Texture.

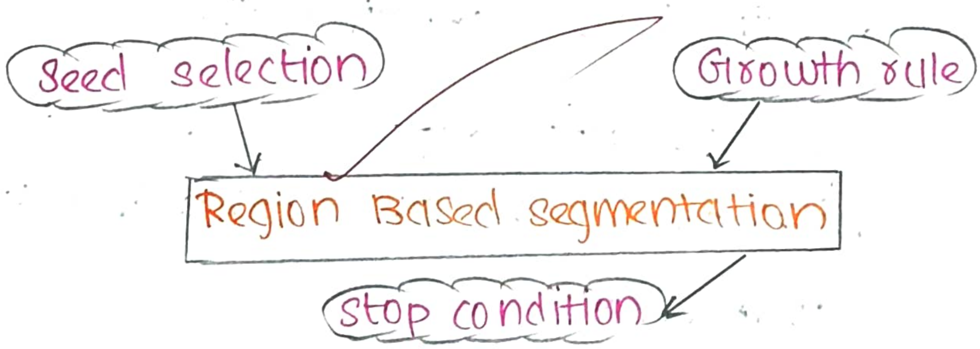
⇒ check whether similar.

⇒ If similar, Add to the region.

⇒ Repeat till No growth

⇓

Texture Based Segmentation Region



To overcome and solve Multiple Regions with similar Intensity but different Texture The suitable Method is "Region based segmentation".

4. An Image Requires edge Enhancement in frequency Domain.

⇒ Edge Enhancement improves object boundaries by emphasizing hidden intensity changes in the image.

⇒ A Highpass Filter (HPF) is used because edges contain High Frequency Information.

⇒ The image is transformed into frequency domain before Applying the filter.

⇒ The HPF suppresses Low-frequency components while preserving and enhancing High frequency components.

⇒ The output Image contains sharper edges. making object detection, feature extraction more efficient.

5

5. A smooth version of an Image is required.

⇒ Image segmentation reduces noise and unwanted Intensity variations present in the Image.

→ A low pass filter is selected it smooths the image by removing High frequencies

→ The image is converted into the frequency domain.

⇒ To get smooth version of Image

↳ The best suitable Method is

" Image segmentation".

⇒ The final Image Appears smooth and. High Frequencies ~~are~~ removed.

⇒ The final Image is suitable for Image Analysis.